Optimization

2.2 Golden section search

José Firmino Aguilar Madeira

Where to place the two points?

In 2.1: f unimodal on $[1, 2]$, $x_1 = 4/3$, $x_2 = 5/3$; since $f(4/3) > f(5/3)$, we kept $[4/3, 2]$.

The point $x_2 = 5/3$ survived, with f already computed — but it lies in the middle of $[4/3, 2]$. If we split into thirds again (1.556 and 1.778), we would not reuse it: two new evaluations per iteration, to reduce the interval to $2/3$.

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We have two interior points and want to reduce the interval many times, reusing one of the evaluations. Where should x_1 and x_2 be?

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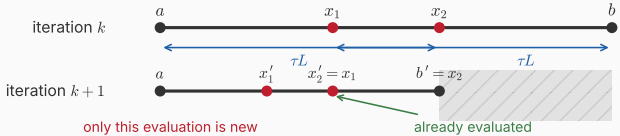
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Two requirements:

1. **Symmetry**: $x_1 = b - \tau L$, $x_2 = a + \tau L$ ($L = b - a$). Whatever the comparison, the new interval has length τL.
2. **Reuse**: the surviving point must be exactly where the next iteration needs one of its two points.

If both hold, each reduction costs a single evaluation of f. Which τ achieves this?

Reusing a point: $\tau^2 = 1 - \tau$



Iteration k: $x_1 = a + (1-\tau)L$, $x_2 = a + \tau L$. Suppose $f(x_1) < f(x_2)$: we keep $[a, x_2]$, of length τL.

At iteration $k+1$ the right-hand point must be at $x_2' = a + \tau(\tau L) = a + \tau^2 L$. For it to be the old x_1:

$$\tau^2 L = (1-\tau)L \quad \Longleftrightarrow \quad \boxed{\tau^2 = 1 - \tau}$$

(the other case, by symmetry, gives the same condition).

$$\tau^2 + \tau - 1 = 0 \;\Rightarrow\; \tau = \frac{\sqrt{5}-1}{2} \approx 0.618.$$

reduction factor $\tau \approx 0.618$
per one new evaluation

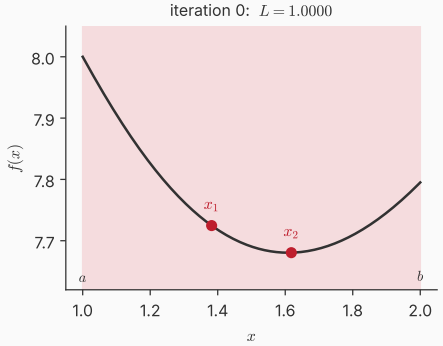
The reciprocal $1/\tau = (1+\sqrt{5})/2 \approx 1.618$ is the *golden ratio*; hence the name of the method.

One iteration by hand

$f(x) = 8e^{1-x} + 7\ln x$ on $[1, 2]$, $L_0 = 1$, $\tau \approx 0.618$:

$$x_1 = 2 - \tau \cdot 1 = 1.382, \qquad x_2 = 1 + \tau \cdot 1 = 1.618,$$

$$f(x_1) = 7.725, \qquad f(x_2) = 7.681.$$



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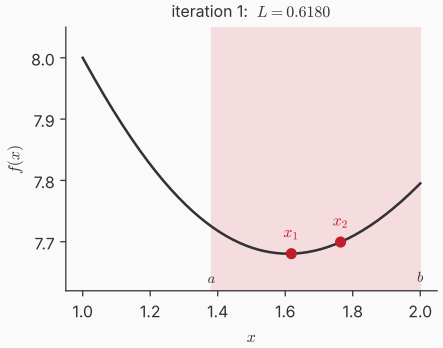
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Since $f(x_1) > f(x_2)$: new interval $[1.382,\ 2]$,
$L_1 = 0.618 = \tau L_0$.

The new left-hand point would be $2 - \tau \cdot 0.618 = 1.618$:
it is the old x_2, already evaluated. We only need

$$x_2' = 1.382 + \tau \cdot 0.618 = 1.764, \qquad f(x_2') = 7.699.$$



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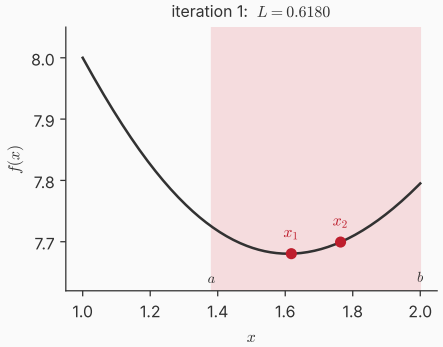
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Thirds (2.1): factor $2/3$ with 2 new evaluations.
Golden section: factor 0.618 with 1 new evaluation.

The algorithm

Golden section on $[a, b]$, tolerance ε_x

1: set $\tau = (\sqrt{5} - 1)/2$; $L = b - a$
2: set $x_1 = b - \tau L$; $x_2 = a + \tau L$ ▷ $x_1 < x_2$
3: set $f_1 = f(x_1)$; $f_2 = f(x_2)$
4: **while** $L > \varepsilon_x$ **do**
5: **if** $f_1 < f_2$ **then** ▷ minimum in $[a, x_2]$
6: set $b = x_2$; $x_2 = x_1$; $f_2 = f_1$
7: set $L = b - a$; $x_1 = b - \tau L$; $f_1 = f(x_1)$
8: **else** ▷ $f_1 \geq f_2$: minimum in $[x_1, b]$
9: set $a = x_1$; $x_1 = x_2$; $f_1 = f_2$
10: set $L = b - a$; $x_2 = a + \tau L$; $f_2 = f(x_2)$
11: **end if**
12: **end while**
13: **return** $x^* \approx (a + b)/2$

- Lines 6 and 9: the surviving point changes role ($x_1 \to x_2$ or $x_2 \to x_1$) and carries its value of f: two evaluations to start, then one per reduction.
- **Tie** $f_1 = f_2$: goes to the *else* branch. This is correct: in that case $x^* \in [x_1, x_2] \subset [x_1, b]$ (2.1). No third branch is needed.
- Without unimodality the guarantee is lost: the comparisons may eliminate the region of the desired minimum.

How many reductions?

Let $L_0 = b - a$ be the initial interval. After k reductions, $L_k = \tau^k L_0$ ($\tau \approx 0.618$) and, taking the midpoint $m_k = (a_k + b_k)/2$ as the estimate, $|m_k - x^*| \leq L_k/2$. With ε_x the tolerance on the interval length (as in the `while` of the algorithm), to guarantee $L_N \leq \varepsilon_x$ — and hence $|m_N - x^*| \leq \varepsilon_x/2$ — it suffices that

$$N \;\geq\; \frac{\ln(\varepsilon_x / L_0)}{\ln \tau} \;=\; \frac{\ln(L_0/\varepsilon_x)}{0.4812}.$$

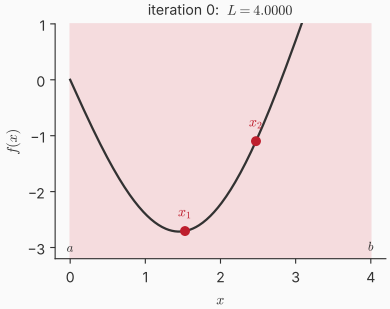
Example

$L_0 = 1$, $\varepsilon_x = 0.01$: $\quad N \geq \ln 100 / 0.4812 = 9.57$, so $N = 10$ reductions ($0.618^{10} = 0.0081$), with $2 + 10 = 12$ evaluations of f.

- Linear convergence with ratio 0.618: each extra decimal digit costs ≈ 4.8 iterations.
- *N does not depend on f* — only on L_0 and ε_x: predictable, but slow compared with Newton (2.5). Bisection on f' reduces by 0.5 per iteration, but needs f'.

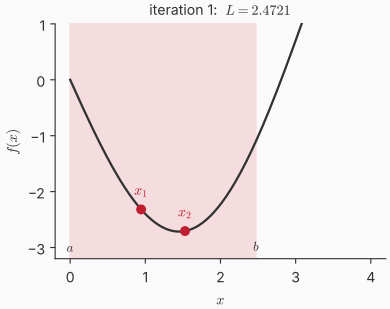
Example 1: $f(x) = -3\sin x + x^2/8$ on $[0, 4]$

k	a	b	x_1	x_2	$f(x_1)$	$f(x_2)$	L_k
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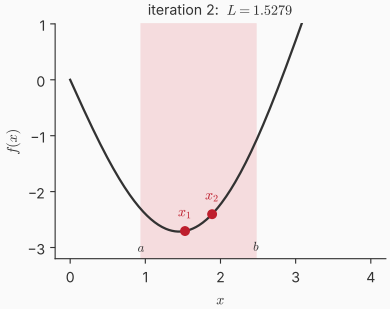
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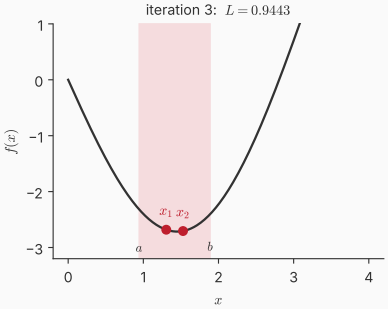
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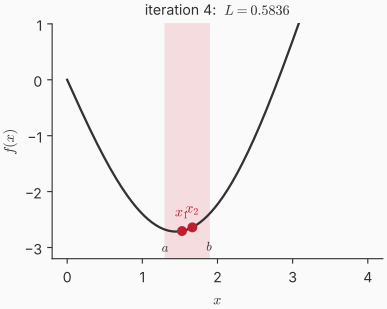
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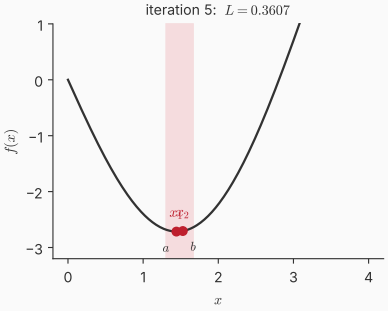
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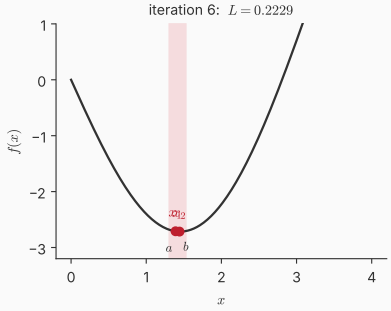
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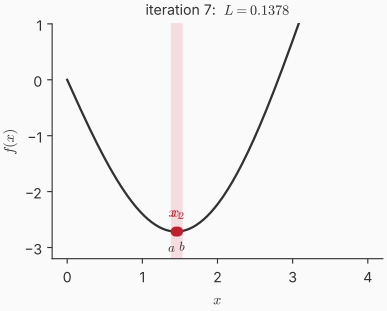
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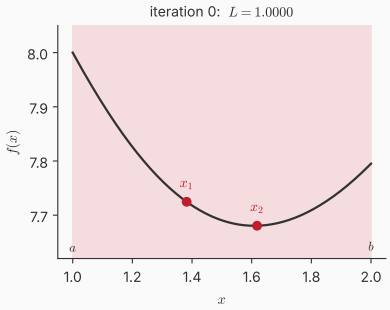
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7	1.3901	1.5279	1.4427	1.4752	−2.7152	−2.7143	0.1378

Row k has $L_k = \tau^k L_0$. After 15 reductions: $L_{15} = 4\tau^{15} = 0.0029$, $x^* \approx 1.4489$, $f(x^*) \approx -2.7153$.



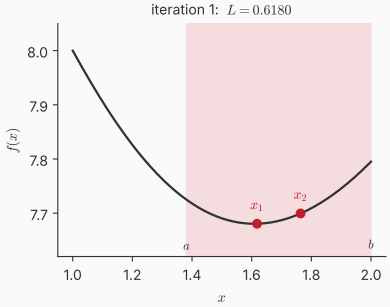
Example 2: $f(x) = 8e^{1-x} + 7\ln x$ on $[1, 2]$ (2019 first test)

k	a	b	x_1	x_2	$f(x_1)$	$f(x_2)$	L_k
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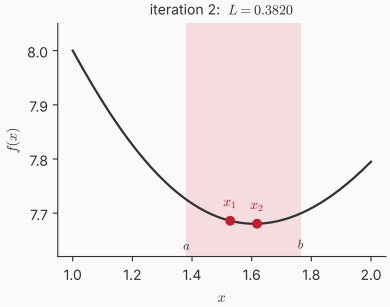
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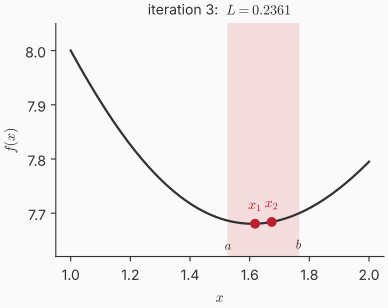
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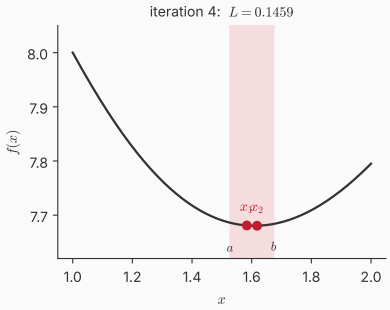
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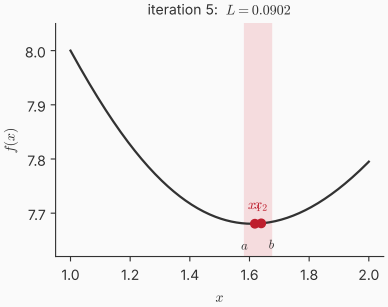
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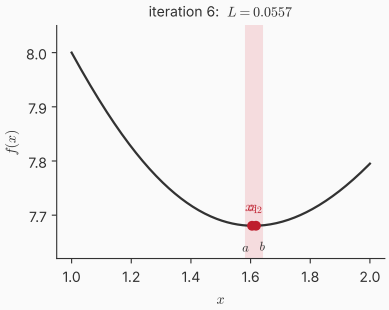
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5	1.5836	1.6738	1.6180	1.6393	7.68051	7.68118	0.0902



Example 2: $f(x) = 8e^{1-x} + 7\ln x$ on $[1, 2]$ (2019 first test)

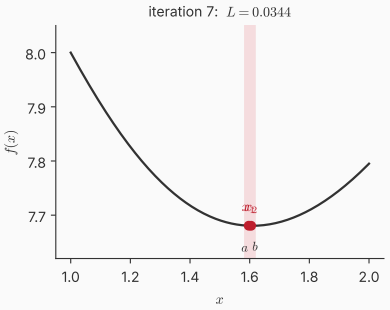
k	a	b	x_1	x_2	$f(x_1)$	$f(x_2)$	L_k
0	1.0000	2.0000	1.3820	1.6180	7.72470	7.68051	1.0000
1	1.3820	2.0000	1.6180	1.7639	7.68051	7.69947	0.6180
2	1.3820	1.7639	1.5279	1.6180	7.68600	7.68051	0.3820
3	1.5279	1.7639	1.6180	1.6738	7.68051	7.68381	0.2361
4	1.5279	1.6738	1.5836	1.6180	7.68100	7.68051	0.1459
5	1.5836	1.6738	1.6180	1.6393	7.68051	7.68118	0.0902
6	1.5836	1.6393	1.6049	1.6180	7.68046	7.68051	0.0557



Example 2: $f(x) = 8e^{1-x} + 7\ln x$ on $[1, 2]$ (2019 first test)

k	a	b	x_1	x_2	$f(x_1)$	$f(x_2)$	L_k
0	1.0000	2.0000	1.3820	1.6180	7.72470	7.68051	1.0000
1	1.3820	2.0000	1.6180	1.7639	7.68051	7.69947	0.6180
2	1.3820	1.7639	1.5279	1.6180	7.68600	7.68051	0.3820
3	1.5279	1.7639	1.6180	1.6738	7.68051	7.68381	0.2361
4	1.5279	1.6738	1.5836	1.6180	7.68100	7.68051	0.1459
5	1.5836	1.6738	1.6180	1.6393	7.68051	7.68118	0.0902
6	1.5836	1.6393	1.6049	1.6180	7.68046	7.68051	0.0557
7	1.5836	1.6180	1.5967	1.6049	7.68058	7.68046	0.0344

After 15 reductions: $L_{15} = \tau^{15} = 0.0007$, $x^* \approx 1.6095$, $f(x^*) \approx 7.68045$. $f'(x) = -8e^{1-x} + 7/x = 0$ has no closed-form solution; to many digits, $x^* = 1.6093811$, $f^* = 7.6804459$. Near the minimum f varies little: f with 5 decimals.



On the computer

The core, in MATLAB (N reductions):

```
tau = (sqrt(5)-1)/2;  L = b - a;
x1 = b - tau*L;  f1 = f(x1);     % x1 < x2
x2 = a + tau*L;  f2 = f(x2);
for k = 1:N
  if f1 < f2                     % minimum in [a,x2]
    b = x2;  x2 = x1;  f2 = f1;
    L = b - a;  x1 = b - tau*L;  f1 = f(x1);
  else                           % minimum in [x1,b]
    a = x1;  x1 = x2;  f1 = f2;
    L = b - a;  x2 = a + tau*L;  f2 = f(x2);
  end
end
x = (a + b)/2;
```

The full function (`GoldenSection`, with `info.nfev` and the table printout) is in the course companion code.

One evaluation per reduction (the other value is reused): $n_f = N + 2$ to locate x^*; if we also evaluate f at the final midpoint, $n_f = N + 3$. So, for $N = 15$, $n_f = 18$.

MATLAB's `fminbnd` combines golden section with parabolic interpolation (2.3).

THE SAME IN PYTHON

```
from scipy.optimize import minimize_scalar
r = minimize_scalar(f, bounds=(0, 4),
                    method='bounded')
r.x, r.fun, r.nfev
```

`bounded` and `fminbnd` use Brent-type strategies; the details may differ. `r.nfev` counts the evaluations.

KEY TAKEAWAYS — 2.2

1. With f unimodal, two interior points allow us to eliminate part of the interval; only the *order* of the values of f is used.
2. Points symmetric at τL from the endpoints, with $\boxed{\tau^2 = 1 - \tau}$, $\tau = (\sqrt{5} - 1)/2 \approx 0.618$: the surviving point is reused, and each reduction costs one evaluation of f.
3. $L_k = \tau^k L_0$ and, with the midpoint m_k, $|m_k - x^*| \leq L_k/2$: linear convergence, with $N \geq \ln(L_0/\varepsilon_x)/0.481$ reductions, independent of f.
4. Robust (needs only unimodality), but slow: each decimal digit costs ≈ 4.8 reductions.

Next (2.3): golden section uses only the *order* of $f(x_1)$ and $f(x_2)$. What if we also use their magnitude — fitting a parabola through three points and jumping to its minimum? — successive quadratic interpolation (Powell's method).

References and resources

References Full list: *Course bibliography*, at the end of the course PDF.

- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Ch. 2.
- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Ch. 3.

Resources

- Companion code: `GoldenSection` (`info.nfev`). MATLAB: `fminbnd`.
- Python: `scipy.optimize.minimize_scalar(f, bounds=(a,b), method='bounded')` (`r.nfev`).